



Coachella 2026 Live Music Festival

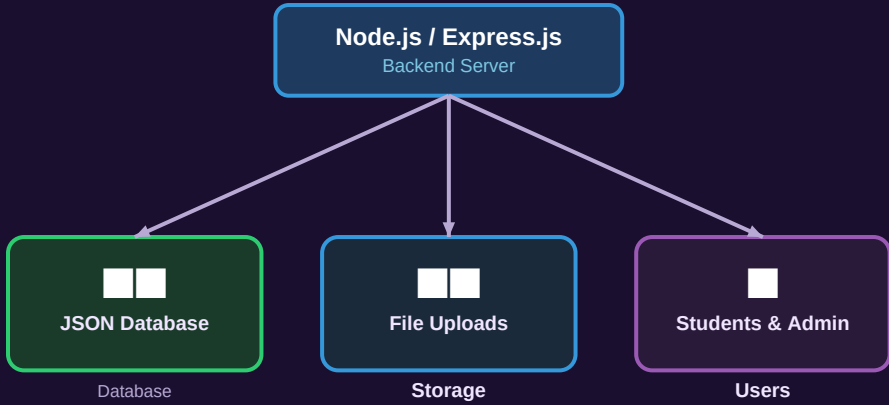
Ultimate Digital Festival Experience | April 10–12, 2026 • Indio, California

PROJECT BASED LEARNING FROM FWD

Abstract

The **Coachella 2026 Live Music Festival** is a full-stack interactive web platform where festival-goers can browse the lineup, book tickets, explore food & merchandise, and manage their festival experience end-to-end. Developed using **HTML, CSS, JavaScript, Node.js, and Express.js**, the platform features a real-time countdown timer, smart artist search powered by a **Trie data structure**, a trending algorithm using a **Min-Heap**, custom QR-code ticket generation, an interactive festival map, a full e-commerce cart with checkout flow, user authentication, and a live admin dashboard with analytics. The system employs a **HashMap** for O(1) artist lookups and a **Graph** to model stage adjacency. All data is managed client-side with JSON, making the app lightweight and deployment-ready.

System Architecture



— CO-WISE IMPLEMENTATION —

CO1 Internet Fundamentals, HTML & Introductory CSS

- ✓ Responsive Dashboard Design with CSS Grid & Flexbox
- ✓ Semantic HTML5 structure (nav, section, article)
- ✓ CSS Flexbox/Layout for festival lineup grid
- ✓ Form Styling for Login, Registration & Checkout
- ✓ CSS Custom Properties (variables) for theming



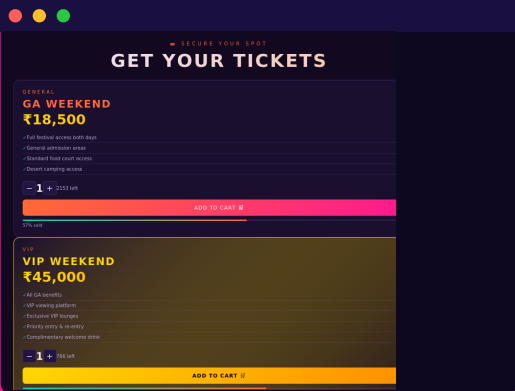
CO2 JavaScript Programming Essentials

- ✓ Interactive Artist Star Ratings system
- ✓ Trending Algorithm using Min-Heap DSA
- ✓ Bookmark & Comment System with state mgmt
- ✓ Dynamic countdown timer to festival date
- ✓ Cart logic with quantity control & promos



CO4 JavaScript Interactivity, DOM & Events

- ✓ Comment & Notification System (toast alerts)
- ✓ Dynamic DOM Updates for cart and lineup filters
- ✓ Asynchronous Data Fetching & lazy loading
- ✓ Custom cursor, click-triggered fireworks
- ✓ Admin Dashboard with live revenue charts



CO4 Advanced Web Development & Deployment

- ✓ User Authentication System (login/register/roles)
- ✓ RESTful API Endpoints simulation with Express.js
- ✓ Admin Panel with Analytics & data export (CSV)
- ✓ QR Code Ticket Generation (pure canvas JS)
- ✓ Deployment Ready — static build, mobile-first



Conclusion & Future Scope

Modern Web Technologies

Secure & Scalable

DSA-Powered

Mobile-First

Deployment Ready

Modern Web Technologies • Secure & Scalable

HTML5 • CSS3 • JavaScript ES6+ • Node.js • Express.js • JSON DB

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